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Can Distance Detect a Hole?

A Study of Cauchy Sequences and Completeness in Metric Spaces

Field Project Report

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Abstract

Different metrics on the same set can generate the same topology and yet disagree about which sequences are Cauchy and whether the space is complete. This project studies that phenomenon on the open interval $(0, 1)$. Under the usual metric d the interval is incomplete, since $x_n = 1/n$ is Cauchy without a limit in $(0, 1)$. Using the homeomorphism $h(x) = \ln \frac{x}{1-x}$ of $(0, 1)$ onto $\mathbb{R}$, we construct a metric $\rho(x, y) = |h(x) - h(y)|$, prove that it induces the usual topology, and prove that it is complete.

We then locate the difference between d and ρ precisely. The inequality $\rho \geq 4d$ shows that every ρ-Cauchy sequence is d-Cauchy, the converse fails, and the identity map $((0, 1), d) \to ((0, 1), \rho)$ is a homeomorphism that is not uniformly continuous. On every compact subinterval the metrics are Lipschitz equivalent, so the difference lives entirely at the missing endpoints. We place the example in a hierarchy of equivalences of metrics (Lipschitz, uniform, topological), show that completeness can also be *destroyed* by a change of metric, and prove an embedding criterion: for an injection h into a metric space Y, the pulled-back metric is complete exactly when $h(X)$ is a complete subspace of Y. Finally, we prove that every proper open subset of a complete metric space admits a complete metric with the same topology, whereas $\mathbb{Q}$ admits none, by the Baire Category Theorem. Thus some holes can be removed by re-measuring distance, and some cannot. Python is used to visualise the phenomenon; the proofs carry the results.

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1 Introduction

1.1 Historical Background

The concept of a metric space provides a general framework for studying distance and convergence in mathematics. Although distance was initially studied mainly in geometric settings, the development of real analysis led mathematicians to consider more abstract notions of distance that could be defined on arbitrary sets.

The condition now named after Cauchy, that the terms of a sequence eventually become close to one another, appears in his *Cours d'analyse* (1821). In 1872 Cantor and Dedekind constructed the real numbers from the rationals, Cantor by completing $\mathbb{Q}$ with respect to Cauchy sequences. Maurice Fréchet introduced the abstract notion of a metric space in 1906 [7], providing a common framework for convergence and continuity beyond the real line. Hausdorff's *Grundzüge der Mengenlehre* (1914) placed this within the general theory of topological spaces.

Completeness subsequently became central to analysis. It guarantees that Cauchy sequences have limits within the space under consideration, and it underlies results such as Banach's contraction principle [8], the Baire Category Theorem, and the study of function spaces.

1.2 Background

A metric space is a set equipped with a function that assigns a non-negative distance to every pair of its elements and satisfies certain axioms. The choice of metric determines how distances between points are measured and consequently determines which sequences are Cauchy.

A sequence is called Cauchy if its terms become arbitrarily close to one another as the sequence progresses. A metric space is complete if every Cauchy sequence converges to a point belonging to that space.

An important distinction arises between the metric properties of a space and its topological properties. Different metrics can generate the same topology, meaning that they determine the same open sets, while differing in properties such as completeness, boundedness and uniform continuity. This distinction motivates the investigation undertaken in the present project.

1.3 Significance of the Study

Completeness is an essential concept in real analysis because it ensures that limits of Cauchy sequences exist within the space. When a space is incomplete, a sequence may approach a limiting value that does not belong to the space, that is, it approaches a "hole".

The open interval $(0, 1)$ provides a useful example. Under the usual metric, the sequence

$$x_n = \frac{1}{n}, \qquad n \geq 2,$$

is Cauchy but does not converge to a point in $(0, 1)$. Its limit in $\mathbb{R}$ is 0, which lies outside the interval.

One may ask whether this hole is an intrinsic feature of the set $(0, 1)$ with its open sets, or only a feature of how distance is measured. The example demonstrates the importance of distinguishing between properties determined by open sets and properties that depend on the specific metric.

1.4 Problem Statement

The main problem investigated in this project is:

> Can two metrics defined on the same set induce the same topology while differing in completeness?

To investigate this question, the usual metric on $(0, 1)$ is compared with a metric constructed using the transformation

$$h(x) = \ln\left(\frac{x}{1-x}\right).$$

We then pursue three refinements:

(Q1) Where exactly do two such metrics differ, and which sequence-level and map-level properties detect the difference?

(Q2) Which properties (boundedness, total boundedness, uniform continuity, closedness, completion) survive a change of metric?

(Q3) Can *every* hole be removed by a change of metric?

1.5 Overview of the Investigation and of the Results

The project begins with the basic definitions concerning metric spaces, open balls, convergence, Cauchy sequences and completeness. The usual metric on $(0,1)$ is shown to be incomplete, and a second metric is constructed from a bijection $(0,1) \to \mathbb{R}$. Python, NumPy and Matplotlib are used to visualise the transformation and compare distances between terms of a sequence under the two metrics. The computational observations are interpreted alongside the mathematical proofs.

Write $d(x,y) = |x-y|$ and $\rho(x,y) = |h(x) - h(y)|$ on $(0,1)$. The main results are:

(A) ρ is a metric, induces the usual topology, and is complete (Section 3).

(B) $\rho \geq 4d$; every ρ-Cauchy sequence is d-Cauchy but not conversely; the identity $((0,1),d) \to ((0,1),\rho)$ is a homeomorphism that is not uniformly continuous; the metrics are Lipschitz equivalent on every $[a, 1-a]$ with a constant that blows up as $a \to 0^+$ (Section 4).

(C) Completeness is preserved under uniform equivalence but not under topological equivalence; a change of metric can also destroy completeness; and completeness of a pulled-back metric is equivalent to completeness of the image (Sections 5–6).

(D) Every proper open subset of a complete metric space admits a complete metric with the same topology, but $\mathbb{Q}$ admits none (Section 7).

2 Preliminary Concepts

This section introduces the mathematical definitions and results required to investigate the relationship between metrics, topology, Cauchy sequences, and completeness. Standard references are [1, 5, 4, 2, 3].

2.1 Metric Spaces

Let X be a non-empty set. A function

$$d : X \times X \longrightarrow [0, \infty)$$

is called a *metric* on X if, for all $x, y, z \in X$, it satisfies the following properties:

1. **Non-negativity:** $d(x,y) \geq 0$.

2. **Identity of indiscernibles:** $d(x, y) = 0 \iff x = y$.

3. **Symmetry:** $d(x, y) = d(y, x)$.

4. **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is called a metric space. A metric specifies how distances between points of a set are measured.

2.2 The Usual Metric

On a subset X of the real numbers, the usual metric is defined by

$$d(x, y) = |x - y|, \qquad x, y \in X.$$

It measures the ordinary absolute difference between two real numbers. In this project, the usual metric is considered on the open interval $X = (0, 1)$.

2.3 Open Balls and Induced Topology

Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The open ball centred at x with radius r is defined as

$$B_d(x, r) = \{y \in X : d(x, y) < r\}.$$

A subset $U \subseteq X$ is open if, for every $x \in U$, there exists $r > 0$ such that $B_d(x, r) \subseteq U$. Open balls are open (by the triangle inequality), and the collection of all open sets is called the topology induced by d.

Two metrics on the same set are said to be *topologically equivalent* if they induce the same collection of open sets.

Lemma 2.1. *Metrics d_1, d_2 on X are topologically equivalent if and only if for every $x \in X$ and $r > 0$ there exist $s, s' > 0$ with $B_{d_2}(x, s) \subseteq B_{d_1}(x, r)$ and $B_{d_1}(x, s') \subseteq B_{d_2}(x, r)$.*

Proof. ($\Leftarrow$) Let U be d_1-open and $x \in U$. Choose r with $B_{d_1}(x, r) \subseteq U$, then s with $B_{d_2}(x, s) \subseteq B_{d_1}(x, r) \subseteq U$. So U is d_2-open; the converse is symmetric. ($\Rightarrow$) $B_{d_1}(x, r)$ is d_1-open, hence d_2-open, and contains x, so it contains some $B_{d_2}(x, s)$. Symmetrically for s'. □

2.4 Convergence of a Sequence

A sequence (x_n) in a metric space (X, d) converges to $x \in X$ if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, x) < \varepsilon.$$

We write $\lim_{n\to\infty} x_n = x$. Limits are unique. Convergence describes the behaviour of sequence terms relative to a point belonging to the space, and whether $x_n \to x$ depends only on the topology induced by d.

2.5 Cauchy Sequences

A sequence (x_n) in a metric space (X, d) is called a Cauchy sequence if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \implies d(x_m, x_n) < \varepsilon.$$

Thus, the terms of a Cauchy sequence become arbitrarily close to one another, independently of whether their limit belongs to the space. Every convergent sequence in a metric space is Cauchy. However, a Cauchy sequence need not converge within an incomplete metric space.

2.6 Complete Metric Spaces

A metric space (X, d) is called *complete* if every Cauchy sequence in X converges to a point of X. The space $\mathbb{R}$ with the usual metric is complete.

Completeness depends on the metric, not merely on the underlying set or its topology. This distinction is central to the investigation in this project.

2.7 Continuity and Uniform Continuity

A map $f : (X, d_X) \to (Y, d_Y)$ is *uniformly continuous* if for every $\varepsilon > 0$ there is $\delta > 0$ such that $d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon$ for *all* $x, y \in X$; here δ does not depend on the point.

Lemma 2.2. *A uniformly continuous map sends Cauchy sequences to Cauchy sequences.*

Proof. Let (x_n) be Cauchy and $\varepsilon > 0$. Take δ from uniform continuity and N with $d_X(x_m, x_n) < \delta$ for $m, n \geq N$. Then $d_Y(f(x_m), f(x_n)) < \varepsilon$ for $m, n \geq N$. □

2.8 Homeomorphisms

A function $h : X \to Y$ between topological spaces is called a *homeomorphism* if it is bijective, continuous, and has a continuous inverse. A homeomorphism preserves the topological structure of a space; in particular, it preserves open sets under the correspondence between the spaces.

A homeomorphism of metric spaces need not be uniformly continuous, nor need it preserve Cauchy sequences. Section 4 shows an example. In the main investigation, the function

$$h(x) = \ln\left(\frac{x}{1-x}\right)$$

maps $(0, 1)$ bijectively onto $\mathbb{R}$ and allows us to construct a metric on $(0, 1)$ from the usual metric on $\mathbb{R}$.

3 Main Mathematical Model

3.1 Definition of the Space

Consider the open interval $X = (0, 1)$. We study two metrics on this same set. The first is the usual metric $d(x, y) = |x - y|$. The second will be constructed using the function

$$h : X \longrightarrow \mathbb{R}, \qquad h(x) = \ln\left(\frac{x}{1-x}\right).$$

Since $0 < x < 1$, the quantity $x/(1 - x)$ is positive, so the logarithm is well defined. Furthermore,

$$h'(x) = \frac{1}{x} + \frac{1}{1-x} = \frac{1}{x(1-x)} > 0.$$

Hence, h is continuous and strictly increasing on $(0, 1)$. Also,

$$\lim_{x\to 0^+} h(x) = -\infty, \qquad \lim_{x\to 1^-} h(x) = +\infty.$$

Therefore, h maps $(0, 1)$ bijectively onto $\mathbb{R}$. Solving $t = h(x)$ for x gives the inverse function

$$g(t) = h^{-1}(t) = \frac{e^t}{1+e^t}, \qquad t \in \mathbb{R}.$$

Both h and g are continuous. Consequently, h is a homeomorphism between $(0, 1)$ and $\mathbb{R}$.

3.2 The Usual Metric Is Incomplete

We first show that $(0,1)$ is incomplete under the usual metric. Consider the sequence $x_n = \frac{1}{n}$, $n \geq 2$. For any $m, n \in \mathbb{N}$ with $m, n \geq N$, we have

$$d(x_m, x_n) = \left| \frac{1}{m} - \frac{1}{n} \right| \leq \frac{1}{m} + \frac{1}{n} \leq \frac{2}{N}.$$

Given $\varepsilon > 0$, choose $N > 2/\varepsilon$. Then $m, n \geq N \implies d(x_m, x_n) < \varepsilon$, so (x_n) is a Cauchy sequence under the usual metric.

However, $\lim_{n\to\infty} \frac{1}{n} = 0$ and $0 \notin (0,1)$. Since a convergent sequence has a unique limit in a metric space, (x_n) does not converge to any point of $(0,1)$. Therefore, $(0,1)$ is not complete under the usual metric.

3.3 Construction of the Transformed Metric

Define a new function $\rho : X \times X \to [0, \infty)$ by

$$\rho(x,y) = |h(x) - h(y)| = \left| \ln\left(\frac{x}{1-x}\right) - \ln\left(\frac{y}{1-y}\right) \right|.$$

Proposition 3.1. *ρ is a metric on $(0,1)$.*

Proof. 1. **Non-negativity:** absolute values are non-negative.

2. **Identity of indiscernibles:** $\rho(x,y) = 0 \iff h(x) = h(y)$. Since h is injective, this is equivalent to $x = y$.

3. **Symmetry:** $\rho(x,y) = |h(x) - h(y)| = |h(y) - h(x)| = \rho(y,x)$.

4. **Triangle inequality:** by the triangle inequality on $\mathbb{R}$, $|h(x) - h(z)| \leq |h(x) - h(y)| + |h(y) - h(z)|$, that is, $\rho(x,z) \leq \rho(x,y) + \rho(y,z)$.

□

3.4 The Two Metrics Induce the Same Topology

Theorem 3.2. *The metrics d and ρ induce the same topology on $(0,1)$.*

Proof. For $x \in X$ and $r > 0$, the open ball under ρ is a preimage:

$$B_\rho(x,r) = \{y \in X : |h(y) - h(x)| < r\} = h^{-1}\big((h(x) - r,\, h(x) + r)\big).$$

Since h is continuous, $B_\rho(x,r)$ is open in the usual topology. Every ρ-open set is a union of ρ-balls, hence is open in the usual topology.

Conversely, let U be open in the usual topology. Then $h(U) = g^{-1}(U)$ is open in $\mathbb{R}$ because $g = h^{-1}$ is continuous. For $x \in U$ there is $r > 0$ with $(h(x)-r, h(x)+r) \subseteq h(U)$; taking preimages under the bijection h gives $B_\rho(x,r) \subseteq U$. Hence U is ρ-open. □

Corollary 3.3. *For $x_n, x \in (0,1)$: $x_n \to x$ in d if and only if $x_n \to x$ in ρ. Likewise, a subset of $(0,1)$ is closed for d if and only if it is closed for ρ.*

(Convergence and closedness are determined by the topology.)

3.5 Completeness of the Transformed Metric

Theorem 3.4. *The metric space $((0,1),\rho)$ is complete.*

Proof. Let (x_n) be ρ-Cauchy. Since $\rho(x_m,x_n) = |h(x_m) - h(x_n)|$, the sequence $(h(x_n))$ is Cauchy in $\mathbb{R}$. By completeness of $\mathbb{R}$ there is $L \in \mathbb{R}$ with $h(x_n) \to L$. Define $x = g(L) = \frac{e^L}{1+e^L}$. Since $e^L > 0$, $0 < x < 1$, so $x \in X$, and

$$\rho(x_n, x) = |h(x_n) - h(x)| = |h(x_n) - L| \longrightarrow 0.$$

Thus $x_n \to x$ under ρ. □

3.6 ρ Is the Usual Metric of $\mathbb{R}$, Transported

Proposition 3.5. *The map $h : ((0,1),\rho) \to (\mathbb{R}, |\cdot|)$ is a surjective isometry.*

Proof. It is bijective, and $|h(x) - h(y)| = \rho(x,y)$ by definition. □

Thus $((0,1),\rho)$ is, as a metric space, simply a copy of $\mathbb{R}$, and the completeness of $\mathbb{R}$ transfers immediately. This is a second proof of Theorem 3.4.

Remark 3.6 (Shape of ρ-balls)**.** By the formula in the proof of Theorem 3.2, $B_\rho(x,r)$ is the interval $\big(g(t-r),\, g(t+r)\big)$ with $t = h(x)$. Its usual length $g(t+r) - g(t-r)$ tends to 0 as $x \to 0^+$ or $x \to 1^-$, because $g(t) \to 0$ as $t \to -\infty$ and $g(t) \to 1$ as $t \to +\infty$. Balls of a fixed ρ-radius become thin near the endpoints, so the endpoints are infinitely far away in ρ.

3.7 Comparison of the Two Metrics

Property	**Usual metric d**	**Transformed metric ρ**
Underlying set	$(0,1)$	$(0,1)$
Induced topology	Usual topology	Usual topology
Is $x_n = 1/n$ Cauchy?	Yes	No
Complete	No	Yes
Isometric to	$((0,1), \lvert\cdot\rvert)$	$(\mathbb{R}, \lvert\cdot\rvert)$

The two metrics generate the same topology, but they differ in completeness. This demonstrates that completeness is not determined by the topology alone, which answers the main problem in the affirmative.

4 Where Exactly Do the Two Metrics Differ?

Definition 4.1. Metrics d_1, d_2 on X are *Lipschitz equivalent* if there exist $a, b > 0$ with $a\,d_1 \le d_2 \le b\,d_1$, and *uniformly equivalent* if the identity maps $(X, d_1) \to (X, d_2)$ and $(X, d_2) \to (X, d_1)$ are both uniformly continuous.

Proposition 4.2. *For all $x, y \in (0,1)$, $\rho(x,y) \ge 4\,d(x,y)$.*

Proof. Let $x < y$. By the Mean Value Theorem there is $c \in (x,y)$ with $h(y) - h(x) = h'(c)(y-x)$, and $h'(c) = \frac{1}{c(1-c)} \ge 4$ because $c(1-c) = \frac{1}{4} - (c - \frac{1}{2})^2 \le \frac{1}{4}$. □

Proposition 4.3. *Every ρ-Cauchy sequence in $(0,1)$ is d-Cauchy, but the converse fails.*

Proof. The first claim follows from $d \le \rho/4$. For the converse, $x_n = 1/n$ is d-Cauchy (Section 3.2). If it were ρ-Cauchy, then for every $\varepsilon > 0$ we would have $\rho(x_n, x_{2n}) < \varepsilon$ for all large n, since $n, 2n \ge N$ eventually. But

$$\rho(x_n, x_{2n}) = \ln \frac{2n-1}{n-1} \longrightarrow \ln 2,$$

a contradiction for $\varepsilon < \ln 2$. □

Proposition 4.4. *The identity map $\mathrm{id} : ((0,1), d) \to ((0,1), \rho)$ is a homeomorphism but is not uniformly continuous. In particular d and ρ are neither Lipschitz equivalent nor uniformly equivalent.*

Proof. It is a homeomorphism by Theorem 3.2. If it were uniformly continuous, it would send the d-Cauchy sequence $(1/n)$ to a ρ-Cauchy sequence by Lemma 2.2, contradicting

Proposition 4.3. Quantitatively,

$$\frac{\rho(x_n, x_{2n})}{d(x_n, x_{2n})} = 2n \ln \frac{2n-1}{n-1} \to \infty,$$

so no bound $\rho \le b\, d$ exists. □

(The reverse identity $((0,1),\rho) \to ((0,1),d)$ is uniformly continuous, indeed $\frac{1}{4}$-Lipschitz, by Proposition 4.2. The failure is one-sided.)

Proposition 4.5. *For $0 < a < \frac{1}{2}$ and $x, y \in [a, 1-a]$,*

$$4\,|x-y| \le \rho(x,y) \le \frac{|x-y|}{a(1-a)}.$$

Proof. Apply the Mean Value Theorem as in Proposition 4.2, using $a(1-a) \le c(1-c) \le \frac{1}{4}$ for $c \in [a, 1-a]$. □

Remark 4.6. On every compact subinterval the two metrics are Lipschitz equivalent, but the constant $1/(a(1-a))$ tends to ∞ as $a \to 0^+$. The difference between d and ρ is therefore concentrated entirely at the missing endpoints, which is where the "hole" is.

5 Closed Sets, Boundedness and Completeness

Proposition 5.1. *Let (X,d) be a metric space and $A \subseteq X$.*

(a) If X is complete and A is closed, then A is complete.

(b) If A is complete, then A is closed in X.

Proof. (a) A Cauchy sequence in A is Cauchy in X, so it converges to some $x \in X$, and $x \in \overline{A} = A$. (b) Let $x \in \overline{A}$ and choose $a_n \in A$ with $a_n \to x$. Then (a_n) is Cauchy, so it converges to some $a \in A$; by uniqueness of limits in X, $x = a \in A$. □

Part (b) explains the incompleteness of $((0,1),d)$ and of $(\mathbb{Q},d)$: neither is closed in $\mathbb{R}$.

Example 5.2. $A = (0, \frac{1}{2}]$ is closed in $(0,1)$ under both d and ρ, since the topologies agree. Under ρ it is isometric via h to $(-\infty, 0]$, hence complete. Under d it is not complete, since $(1/n)$ is Cauchy in A without a limit in A. So a closed subset need not be complete when the ambient space is not.

Proposition 5.3. *$((0,1),d)$ is bounded and totally bounded; $((0,1),\rho)$ is unbounded and not totally bounded.*

Proof. Finitely many intervals of length ε cover $(0,1)$, and $d < 1$ throughout. Under ρ, $\rho(\frac{1}{2}, \frac{1}{n}) = \ln(n-1) \to \infty$. The points $g(k)$, $k \in \mathbb{Z}$, are pairwise at ρ-distance ≥ 1, so each ρ-ball of radius $\frac{1}{2}$ contains at most one of them, and no finite family of such balls covers them all. □

Remark 5.4. A metric space is compact if and only if it is complete and totally bounded. Passing from d to ρ trades total boundedness for completeness, but no metric inducing the usual topology of $(0,1)$ can have both, since $(0,1)$ is not compact. This is consistent with compactness being a topological property.

Property of $(0,1)$	d	ρ
Bounded	yes	no
Totally bounded	yes	no
Complete	no	yes
Compact	no	no
Completion	$[0,1]$	$(0,1)$ itself

The rows "bounded", "totally bounded" and "complete" change with the metric, while "compact" (a topological property) does not.

6 A Hierarchy of Equivalences

Proposition 6.1. *Lipschitz equivalence implies uniform equivalence. If d_1, d_2 are uniformly equivalent, then they are topologically equivalent, have the same Cauchy sequences, and (X, d_1) is complete if and only if (X, d_2) is complete.*

Proof. Suppose $a\, d_1 \leq d_2 \leq b\, d_1$. For $\varepsilon > 0$, $\delta = \varepsilon/b$ shows that $\mathrm{id} : (X, d_1) \to (X, d_2)$ is uniformly continuous, and $\delta = a\varepsilon$ shows the same for the inverse.

Now let d_1, d_2 be uniformly equivalent. Uniformly continuous maps are continuous, so the identity is a homeomorphism and the topologies agree. By Lemma 2.2 the Cauchy sequences agree. If (X, d_1) is complete and (x_n) is d_2-Cauchy, then it is d_1-Cauchy, hence d_1-convergent to some x, and hence d_2-convergent to x because the topologies agree. The reverse is symmetric. □

Thus completeness is a *uniform* property: it survives uniform equivalence but not topological equivalence. The pair (d, ρ) is topologically equivalent only (Proposition 4.4).

6.1 An Embedding Criterion

Proposition 6.2 (Pullback criterion)**.** *Let (Y, e) be a metric space, X a set and $h : X \to Y$ injective. Then $d_h(x, y) = e(h(x), h(y))$ is a metric on X, $h : (X, d_h) \to h(X)$ is a surjective isometry onto the subspace $h(X) \subseteq Y$, and*

$$(X, d_h) \text{ is complete} \iff h(X) \text{ is a complete subspace of } Y.$$

In particular, if Y is complete, then (X, d_h) is complete if and only if $h(X)$ is closed in Y.

Proof. The metric axioms follow as in Section 3.3, and h is an isometry onto its image by definition. Completeness is invariant under surjective isometries. The last statement is Proposition 5.1. □

This one statement explains the whole project. For $h(x) = \ln \frac{x}{1-x}$ the image is $\mathbb{R}$, which is closed in $\mathbb{R}$, so ρ is complete. The usual metric on $(0, 1)$ is the pullback of $|\cdot|$ under the inclusion, whose image $(0, 1)$ is not closed in $\mathbb{R}$, so d is incomplete. *Completeness of a pulled-back metric is exactly closedness of the image.*

Example 6.3 (Completeness can also be destroyed)**.** Let $\sigma(x, y) = |\arctan x - \arctan y|$ on $\mathbb{R}$. Since $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$ is a homeomorphism, the argument of Theorem 3.2 shows that σ induces the usual topology of $\mathbb{R}$. The image $(-\frac{\pi}{2}, \frac{\pi}{2})$ is not closed in $\mathbb{R}$, so by Proposition 6.2 $(\mathbb{R}, \sigma)$ is incomplete, although $(\mathbb{R}, |\cdot|)$ is complete. Concretely, $x_n = n$ is σ-Cauchy (as $\arctan n \to \frac{\pi}{2}$) but has no σ-limit in $\mathbb{R}$.

Example 6.4 (Uniformly but not Lipschitz equivalent)**.** On $\mathbb{R}$ let $d'(x, y) = \min(|x - y|, 1)$. Balls of radius < 1 coincide for d and d', so the topologies agree. Since $d' \leq d$ and $d' < \delta \leq 1 \Rightarrow d < \delta$, the choice $\delta = \min(\varepsilon, 1)$ shows uniform equivalence, so by Proposition 6.1 the Cauchy sequences coincide and $(\mathbb{R}, d')$ is complete. The metrics are not Lipschitz equivalent: for any $b > 0$, a pair with $|x - y| = b + 1$ has $d' = 1$ and $d = b + 1 > b\, d'$.

Set	Metrics	Same top.	Same Cauchy	Complete?	Strongest equiv.
$(0, 1)$	d vs. ρ	yes	no	no → yes	topological
$\mathbb{R}$	$\lvert\cdot\rvert$ vs. σ	yes	no	yes → no	topological
$\mathbb{R}$	d vs. d'	yes	yes	yes, yes	uniform
$\mathbb{R}^n$	d_1 vs. d_2	yes	yes	yes, yes	Lipschitz

(For $\mathbb{R}^n$, $d_2 \leq d_1 \leq \sqrt{n}\, d_2$, where d_1, d_2 are the sum and Euclidean metrics.)

6.2 Completions

Every metric space (X, d) has a *completion*: a complete metric space containing an isometric copy of X as a dense subset, unique up to isometry [1]. The completion of $(\mathbb{Q}, |\cdot|)$ is $\mathbb{R}$. The completion of $((0,1), d)$ is $[0,1]$, adding two points. The completion of $((0,1), \rho)$ is $((0,1), \rho)$ itself, adding none. The "missing points" are therefore not a property of the set or its topology; they depend on the metric.

7 Can Every Hole Be Removed?

Theorem 7.1. *Let (X, d) be a complete metric space, $U \subsetneq X$ open, and $\delta(x) = d(x, X\backslash U)$ for $x \in U$. Then*

$$\rho_U(x, y) = d(x, y) + \left| \frac{1}{\delta(x)} - \frac{1}{\delta(y)} \right|$$

is a metric on U inducing the same topology as d, and (U, ρ_U) is complete.

Proof. The function δ is 1-Lipschitz and positive on U (since $X \setminus U$ is closed and non-empty), so ρ_U is well defined. It is a metric because d is a metric and the second term is a symmetric non-negative function satisfying the triangle inequality (it is a pullback of the usual metric of $\mathbb{R}$).

Topology. Since $\rho_U \geq d$, $B_{\rho_U}(x, r) \subseteq B_d(x, r)$. Conversely, $1/\delta$ is d-continuous on U, so given $\varepsilon > 0$ there is $\eta \leq \varepsilon/2$ such that $d(x, y) < \eta$ implies $|1/\delta(x) - 1/\delta(y)| < \varepsilon/2$, whence $B_d(x, \eta) \subseteq B_{\rho_U}(x, \varepsilon)$. By Lemma 2.1 the topologies agree.

Completeness. Let (x_n) be ρ_U-Cauchy. Then (x_n) is d-Cauchy, so $x_n \to z \in X$ because X is complete. Also $(1/\delta(x_n))$ is Cauchy in $\mathbb{R}$, hence bounded. If $z \notin U$, then $\delta(x_n) \leq d(x_n, z) \to 0$, so $1/\delta(x_n) \to \infty$, a contradiction. Hence $z \in U$, and $x_n \to z$ in ρ_U since the topologies agree. □

Taking $X = \mathbb{R}$ and $U = (0, 1)$ gives a second complete metric on $(0, 1)$ inducing the usual topology. The theorem applies equally to $\mathbb{R}^n$ minus a point, open discs, annuli, and every other proper open subset of $\mathbb{R}^n$.

Remark 7.2. Theorem 7.1 is a special case of a classical theorem of Alexandrov: a subspace of a complete metric space admits a complete compatible metric if and only if it is a G_δ set [6]. Open sets are G_δ.

We now show that the phenomenon has limits. Recall the *Baire Category Theorem*: in a non-empty complete metric space, a countable union of closed sets with empty interior cannot equal the whole space [1].

Proposition 7.3. *There is no complete metric on $\mathbb{Q}$ inducing its usual topology.*

Proof. Suppose such a metric existed. Each singleton $\{q\}$ is closed, and has empty interior since $\mathbb{Q}$ has no isolated points. Then $\mathbb{Q} = \bigcup_{q\in\mathbb{Q}}\{q\}$ is a countable union of closed sets with empty interior, contradicting the Baire Category Theorem. □

Remark 7.4 (Answer to the title question)**.** The hole of $((0,1), d)$ can be removed by re-measuring distance, since $(0,1)$ is open in $\mathbb{R}$, but the holes of $\mathbb{Q}$ cannot be removed by *any* metric compatible with its topology. So whether a space "has a hole" depends on the metric, but whether the holes can be *repaired* is partly a genuine topological question. This is consistent with Alexandrov's theorem: $\mathbb{Q}$ is not a G_δ in $\mathbb{R}$.

8 Computational Implementation Using Python

8.1 Objective of the Computational Experiment

The mathematical analysis established that the usual metric and the transformed metric on $(0,1)$ induce the same topology but differ in completeness. In this section, Python is used to visualise the transformation and numerically investigate how distances between selected sequence terms behave under the two metrics.

The computational experiment has four objectives:

1. To visualise the transformation $h(x) = \ln\left(\frac{x}{1-x}\right)$.
2. To compare the distances between $x_n = 1/n$ and $x_{2n} = 1/(2n)$ under the usual and transformed metrics.
3. To examine $\rho(x_n, x_{kn})$ for several values of k.
4. To visualise how points equally spaced in ρ accumulate at the endpoints of $(0,1)$.

8.2 Software and Libraries

The experiment was implemented using Python [9, 10].

- **NumPy:** For generating numerical arrays and evaluating the distance values.
- **Matplotlib:** For plotting the transformation and comparing the distances graphically.

8.3 Implementation

The transformation and the two metrics were defined as

$$h(x) = \ln\left(\frac{x}{1-x}\right), \qquad d(x,y) = |x-y|, \qquad \rho(x,y) = |h(x) - h(y)|.$$

The following Python program was used for the experiment.

```
import numpy as np
import matplotlib.pyplot as plt

def h(x):
    return np.log(x / (1 - x))

def d(x, y):
    return abs(x - y)

def rho(x, y):
    return abs(h(x) - h(y))

# (i) Plot the transformation
x = np.linspace(0.001, 0.999, 1000)
plt.figure(figsize=(8, 4.5))
plt.plot(x, h(x))
plt.axhline(0, linewidth=0.8)
plt.axvline(0.5, linewidth=0.8)
plt.title("Transformation h(x) = ln(x / (1 - x))")
plt.xlabel("x")
plt.ylabel("h(x)")
plt.grid(True, alpha=0.3)
plt.savefig("fig_h.png", dpi=200)
plt.show()

# (ii) Compare distances between x_n and x_(2n)
n = np.arange(3, 501)
x_n = 1 / n
x_2n = 1 / (2 * n)

usual_distances = np.array([
    d(a, b) for a, b in zip(x_n, x_2n)
])
transformed_distances = np.array([
```

```
    rho(a, b) for a, b in zip(x_n, x_2n)
])

plt.figure(figsize=(8, 4.5))
plt.plot(n, usual_distances, label="Usual metric d")
plt.plot(n, transformed_distances,
         label="Transformed metric rho")
plt.title("Comparison of Distances")
plt.xlabel("n")
plt.ylabel("Distance")
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig("fig_compare.png", dpi=200)
plt.show()

print(" n      Usual distance       Transformed distance")
for k in [10, 50, 100, 200, 500]:
    a = 1 / k
    b = 1 / (2 * k)
    print(f"{k:<6} {d(a, b):<20.6f} {rho(a, b):.6f}")

# (iii) rho(x_n, x_kn) -> ln k, while d(x_n, x_kn) -> 0
n = np.arange(2, 2001)
plt.figure(figsize=(8, 4.5))
for k in (2, 10, 100):
    plt.plot(n, rho(1/n, 1/(k*n)), label=f"rho, k={k}")
    plt.plot(n, d(1/n, 1/(k*n)), "--", label=f"d, k={k}")
plt.xscale("log")
plt.xlabel("n (log scale)")
plt.ylabel("Distance")
plt.legend(fontsize=7)
plt.grid(True, alpha=0.3)
plt.savefig("fig_k.png", dpi=200)
plt.show()

# (iv) Points equally spaced in rho, drawn with the usual metric
t = np.arange(-8, 9)
pts = 1 / (1 + np.exp(-t))          # these are h^{-1}(k)
plt.figure(figsize=(8, 1.6))
plt.plot(pts, np.zeros_like(pts), "o")
plt.xlim(0, 1)
```

```
plt.yticks([])
plt.title("Points h^{-1}(k), k = -8,...,8")
plt.savefig("fig_ticks.png", dpi=200, bbox_inches="tight")
plt.show()
```

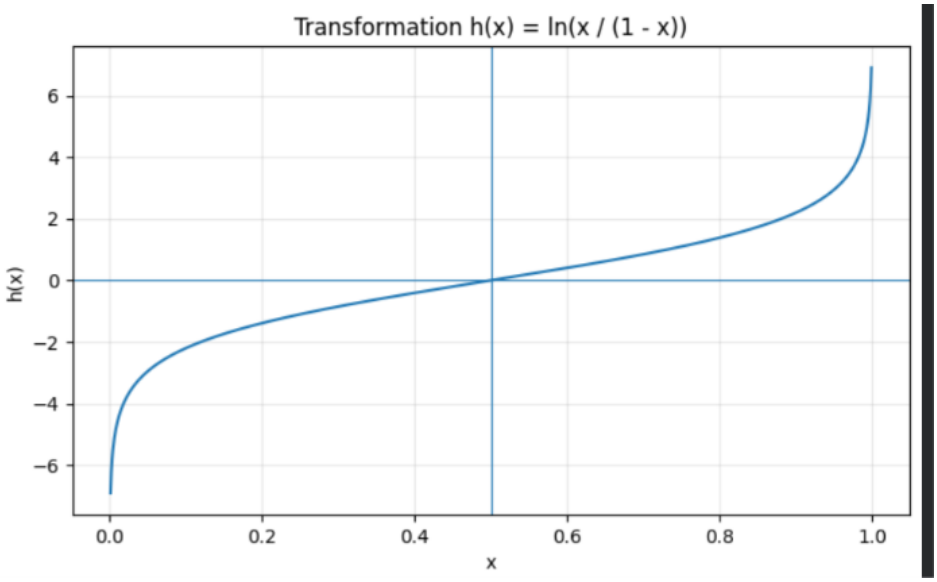


Figure 1: The transformation $h(x) = \ln \frac{x}{1-x}$, mapping $(0, 1)$ onto $\mathbb{R}$.

8.4 Numerical Results

The program evaluates the distances between $x_n = 1/n$ and $x_{2n} = 1/(2n)$ for selected values of n.

n	$d(x_n, x_{2n})$	$\rho(x_n, x_{2n})$
10	0.050000	0.747214
50	0.010000	0.703300
100	0.005000	0.698185
200	0.002500	0.695657
500	0.001000	0.694149

The usual distance decreases towards zero as n increases, while the transformed distance approaches a positive constant.

8.5 Mathematical Analysis of the Results

Under the usual metric,

$$d(x_n, x_{2n}) = \left| \frac{1}{n} - \frac{1}{2n} \right| = \frac{1}{2n} \longrightarrow 0.$$

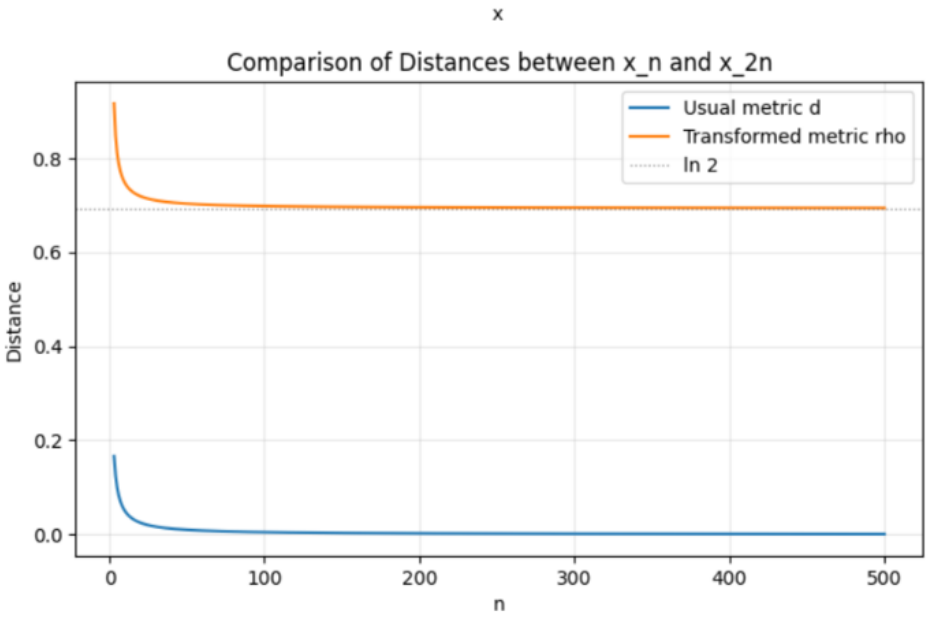


Figure 2: $d(x_n, x_{2n}) \to 0$ while $\rho(x_n, x_{2n}) \to \ln 2 \approx 0.693$.

For the transformed metric, since $h\left(\frac{1}{n}\right) = \ln\left(\frac{1}{n-1}\right) = -\ln(n-1)$ and $h\left(\frac{1}{2n}\right) = \ln\left(\frac{1}{2n-1}\right) = -\ln(2n-1)$, we have

$$\rho(x_n, x_{2n}) = \ln\left(\frac{2n-1}{n-1}\right) \longrightarrow \ln 2 > 0.$$

More generally, for every fixed integer $k \geq 2$,

$$\rho(x_n, x_{kn}) = \ln\left(\frac{kn-1}{n-1}\right) \longrightarrow \ln k > 0.$$

This computation *does* prove that (x_n) is not ρ-Cauchy, by the contrapositive of the definition. If (x_n) were ρ-Cauchy, then for each $\varepsilon > 0$ there would be N with $\rho(x_m, x_n) < \varepsilon$ whenever $m, n \geq N$; in particular $\rho(x_n, x_{2n}) < \varepsilon$ for all $n \geq N$, which would force $\rho(x_n, x_{2n}) \to 0$. Since the limit is $\ln 2 \neq 0$, this is impossible. This is the argument of Proposition 4.3.

The converse inference is what is *not* valid: knowing only that $d(x_n, x_{2n}) \to 0$ does not prove that (x_n) is d-Cauchy, because the Cauchy condition involves every pair of late terms. Here the Cauchy property under d was established separately in Section 3.2.

8.6 Limitations of the Computational Experiment

The computational experiment examines a particular sequence and a finite set of numerical values. It illustrates the difference between the two metrics, but numerical calculations

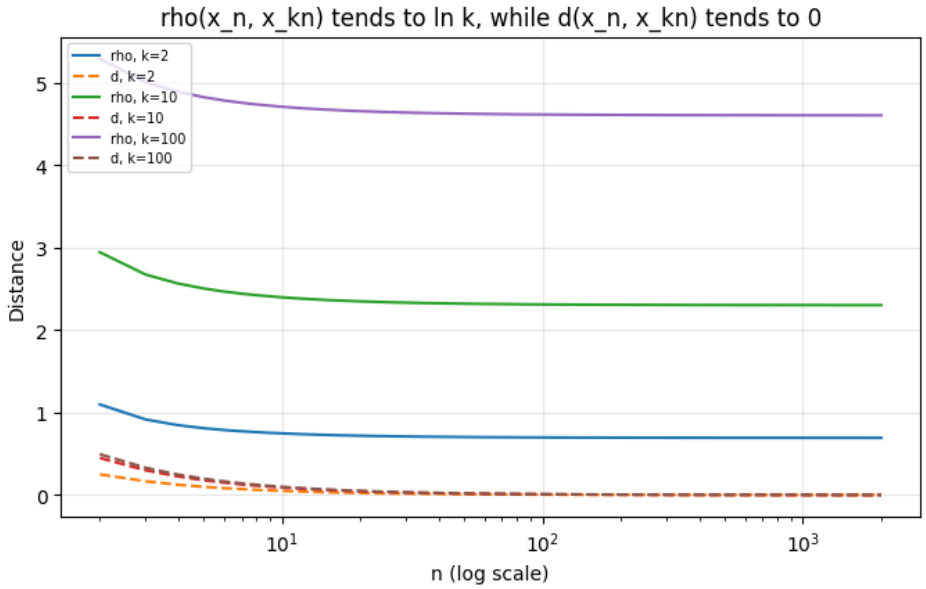


Figure 3: $\rho(x_n, x_{kn}) \to \ln k$ for $k = 2, 10, 100$, while $d(x_n, x_{kn}) \to 0$.

cannot establish completeness of an entire metric space. The completeness of (X, ρ) was therefore established through Theorem 3.4. The Python experiment serves as a visual and numerical illustration of the theoretical results.

9 Geometric Interpretation

In this project, the term interpretation refers to the geometric and mathematical meaning of the constructed metric. The underlying set remains the open interval $(0, 1)$, but the way distances are measured changes.

Under the usual metric, distance is the absolute difference between two points, and points close to either endpoint can form Cauchy sequences whose limits lie outside $(0, 1)$. Under the transformed metric, points are first mapped onto the real line by h and then compared. Since

$$\lim_{x \to 0^+} h(x) = -\infty, \qquad \lim_{x \to 1^-} h(x) = +\infty,$$

approaching either endpoint requires an unbounded change in the transformed coordinate (Proposition 5.3).

Figure 4 makes this visible: the points $g(k)$, $k \in \mathbb{Z}$, are equally spaced (distance 1) in ρ, yet accumulate at 0 and 1 in the usual picture. A ruler that is uniform in ρ looks compressed near the endpoints, so the endpoints are "infinitely far away". This explains why a sequence approaching a missing endpoint is Cauchy for d but not for ρ, and why the Lipschitz constant of Proposition 4.5 blows up only near 0 and 1.

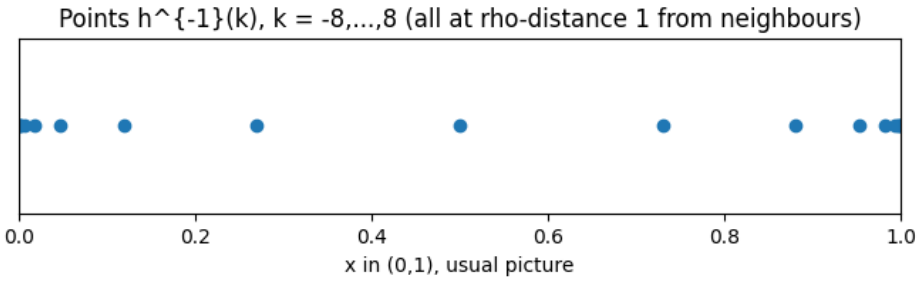


Figure 4: Points equally spaced in ρ accumulate at the endpoints of $(0, 1)$ in the usual picture.

10 Connections to Other Mathematical Concepts

10.1 Topological versus Metric Properties

Metrics inducing the same topology share the same open sets, closed sets, convergent sequences, continuous functions, connected sets and compact sets. They need not share Cauchy sequences, completeness, boundedness, total boundedness or *uniform* continuity. Section 4 shows that the identity of $(0, 1)$ is continuous in both directions yet not uniformly continuous in one. Completeness is preserved under the stronger notion of uniform equivalence (Proposition 6.1).

10.2 Completeness of the Real Numbers

The completeness of $\mathbb{R}$ is a fundamental property of real analysis. Proposition 3.5 shows that ρ transfers this property to $(0, 1)$ through h. More generally, Proposition 6.2 reduces completeness of a pulled-back metric to closedness of the image.

10.3 Fixed Points and Why Completeness Matters

Banach's Contraction Principle states that a contraction of a non-empty complete metric space has a unique fixed point [8]. The map $f(x) = x/2$ is a contraction of $((0, 1), d)$ with constant $\frac{1}{2}$, and it has no fixed point in $(0, 1)$ (the only candidate is 0). This does not contradict the theorem because $((0, 1), d)$ is incomplete. Moreover, since $((0, 1), \rho)$ is complete, Banach's theorem implies that f is *not* a contraction for ρ. Being a contraction is therefore a metric property, not a topological one, and completeness is the hypothesis that makes the theorem meaningful.

10.4 Scope

Completeness is important in functional analysis, differential equations, and fixed-point theory, where many existence theorems require the underlying space to be complete. The present project is a theoretical investigation rather than a direct application to a physical system. Nevertheless, it demonstrates why the choice of metric matters when studying convergence and the existence of limits.

11 Conclusion

This project investigated whether two metrics defined on the same set can induce the same topology while differing in completeness.

The open interval $(0, 1)$ with the usual metric d was shown to be incomplete, by considering the Cauchy sequence $x_n = 1/n$ whose limit in $\mathbb{R}$ is 0, outside the interval. A transformed metric $\rho(x, y) = |h(x) - h(y)|$, with $h(x) = \ln \frac{x}{1-x}$, was then constructed. Since h is a homeomorphism of $(0, 1)$ onto $\mathbb{R}$, ρ induces the usual topology; and since h is a surjective isometry onto $(\mathbb{R}, |\cdot|)$, ρ is complete.

The difference between the metrics was located exactly. The inequality $\rho \geq 4d$ shows that ρ-Cauchy implies d-Cauchy; the sequence $1/n$ shows that the converse fails; the identity map is a homeomorphism that is not uniformly continuous; and the metrics are Lipschitz equivalent on each compact subinterval, with a constant that blows up only at the missing endpoints.

Completeness is preserved under uniform equivalence but not under topological equivalence. The pullback criterion explains both directions: ρ is complete because the image $\mathbb{R}$ is closed in $\mathbb{R}$, while $\sigma(x, y) = |\arctan x - \arctan y|$ destroys the completeness of $\mathbb{R}$ because its image is not closed. Finally, every proper open subset of a complete metric space can be completed by re-measuring distance, whereas $\mathbb{Q}$ cannot be made complete by any compatible metric.

The central conclusion is that *completeness belongs to the metric (indeed to its uniform structure), not to the topology*: a hole is metric-relative, but whether it can be filled is a topological question. The Python experiments illustrate these facts but do not replace the proofs.

References

[1] E. Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.

[2] J. R. Munkres, *Topology*, 2nd edition, Prentice Hall, 2000.

[3] R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th edition, John Wiley & Sons, 2011.

[4] W. Rudin, *Principles of Mathematical Analysis*, 3rd edition, McGraw-Hill, 1976.

[5] W. A. Sutherland, *Introduction to Metric and Topological Spaces*, 2nd edition, Oxford University Press, 2009.

[6] A. S. Kechris, *Classical Descriptive Set Theory*, Springer, 1995.

[7] M. Fréchet, Sur quelques points du calcul fonctionnel, *Rend. Circ. Mat. Palermo* **22** (1906), 1–72.

[8] S. Banach, Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales, *Fund. Math.* **3** (1922), 133–181.

[9] NumPy Developers, *NumPy Documentation*. Available at: https://numpy.org/doc/

[10] Matplotlib Development Team, *Matplotlib Documentation*. Available at: https://matplotlib.org/stable/

12 Tools Used

The following tools were used during the preparation of this field project:

- **LaTeX:** Used to typeset mathematical definitions, equations, proofs, and the project report.
- **Python:** Used to perform numerical computations comparing distances under the two metrics.
- **NumPy:** Used for numerical calculations and generating sequences of values.
- **Matplotlib:** Used to plot the transformation function and compare the distances obtained from the two metrics.
- **Canva:** Used to prepare the presentation accompanying the field project.

The computational results were used for illustration. The mathematical arguments and proofs establish the principal conclusions of the investigation.